Energy Correlators in V+X
as a Benchmark Observable for Precision QCD

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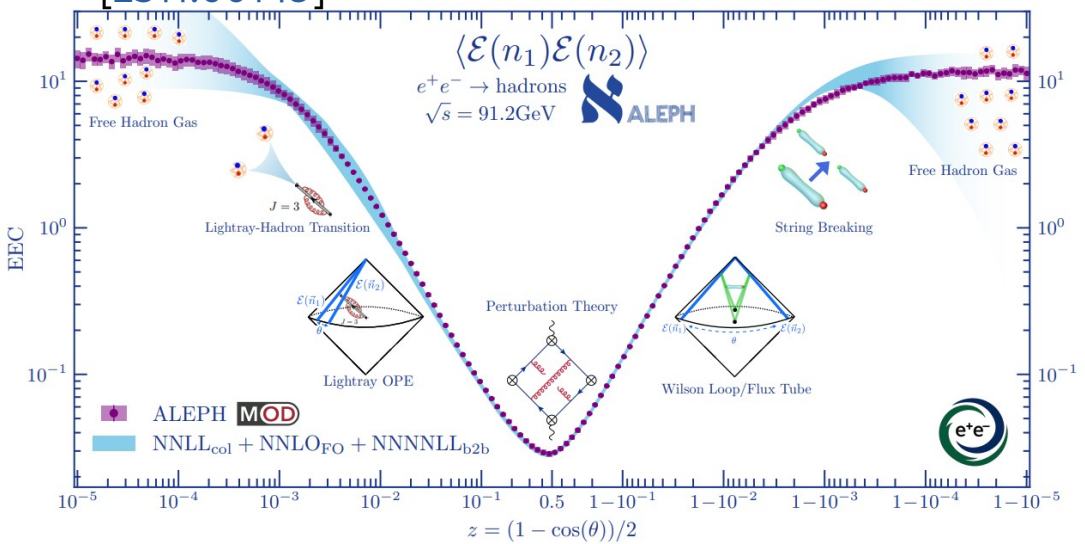
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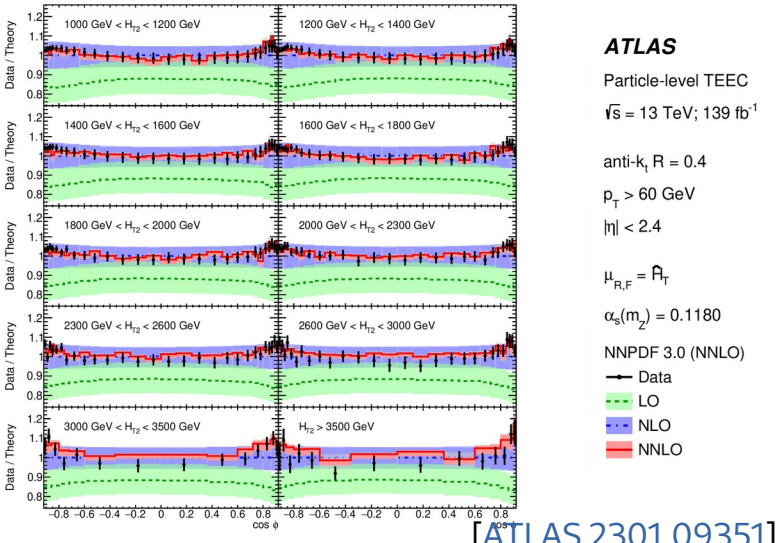
Energy-energy correlators as probe for QCD

e+e- alliance: https://ee-alliance.org

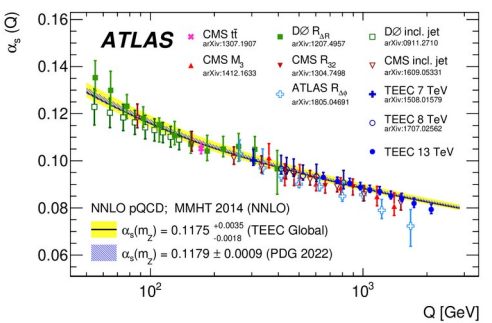
[2511.00149]



Jet-based TEEC



[ATLAS 2301.09351]



Projected Energy-Energy Correlators

Energy flow operator:
[hep-ph/9512370][hep-ph/9601308][0803.1467]

$$\mathcal{E}(\vec{n}) = \int_0^\infty dt \lim_{r\to\infty} r^2\, n^i\, T_{0i}(t, r\vec{n})$$

idealised **"calorimeter"**

Originally defined for e^+e^-:
[Basham, L. Brown, S. Ellis, and S. Love'79][2004.11381]

$$\frac{d\sigma^{[N]}_{e^+e^-}}{dx_L} = \int \prod_{i=1}^{N} d\Omega_{\vec{n}_i}\, \delta\Big(x_L - \max_{1\le i<j\le N} x_{ij}\Big) \underbrace{\int d^4x\, \frac{e^{iq\cdot x}}{Q^N}\, \langle 0|\mathcal{O}^\dagger(x)\, \mathcal{E}(\vec{n}_1)\cdots\mathcal{E}(\vec{n}_N)\, \mathcal{O}(0)|0\rangle}$$

classic energy-energy correlator

projected N -point correlators (ENC):
integrate out angular dependence expect for largest angle

In perturbation theory:

$$\frac{d\sigma^{[N]}_{e^+e^-}}{dx_L} = \frac{1}{2Q^2} \sum_m \int d\Phi_m \sum_{X_m} |\mathcal{M}_{X_m}|^2 \times \sum_{1\le i_1,\ldots,i_N\le m} \frac{\prod_{a=1}^N E_{i_a}}{Q^N}\, \delta\big(x_L - \max\{x_{i_1 i_2}, x_{i_1 i_3}, \ldots, x_{i_{N-1} i_N}\}\big)$$

Energy-energy correlators at hadron colliders

As a jet substructure observable:
[2205.03414]

sum over all particles in the jet

$$\frac{d\sigma_{\text{jet}}^{[N]}}{dR_L} = \int d\sigma_{pp\to\text{jet}+X} \sum_{i_1,\ldots,i_N\in\text{jet}} \frac{\prod_{a=1}^{N} p_{i_a,T}}{(p_{\text{jet},T})^N} \delta\big(R_L - \max\{R_{i_1 i_2}, R_{i_1 i_3}, \ldots, R_{i_{N-1} i_N}\}\big)$$

introduces additional dependence on jet algorithm, R
→ additonal scale $p_T \cdot R$
→ normalisation is more tricky (not the hard scattering scale → additional uncertainties)
→ limits the angular range: potential cluster and non-global logarithms

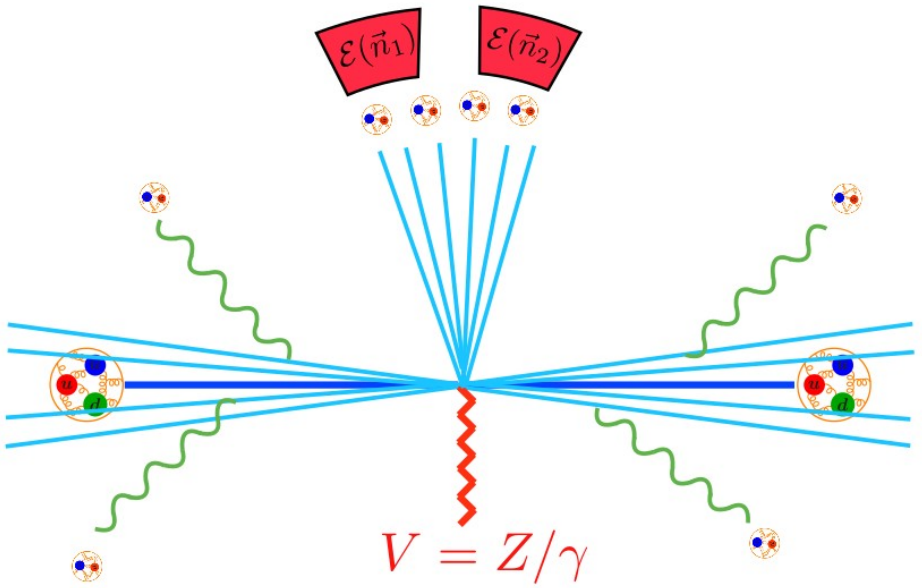
Using a hard tag resolves this:
(see also [2602.03180])

$$\frac{d\sigma^{[N]}}{dR_L} = \sum_m \sum_{X_m} \int d\sigma_{pp\to V+X_m} \times \sum_{1\le i_1,\ldots,i_N\le m} \frac{\prod_{a=1}^{N} p_{i_a,T}}{p_{V,T}^N} \delta\big(R_L - \max\{R_{i_1 i_2}, R_{i_1 i_3}, \ldots, R_{i_{N-1} i_N}\}\big)$$

The setup

Measure the correlators on whole final state

(in our case all particles, but could also be tracks)

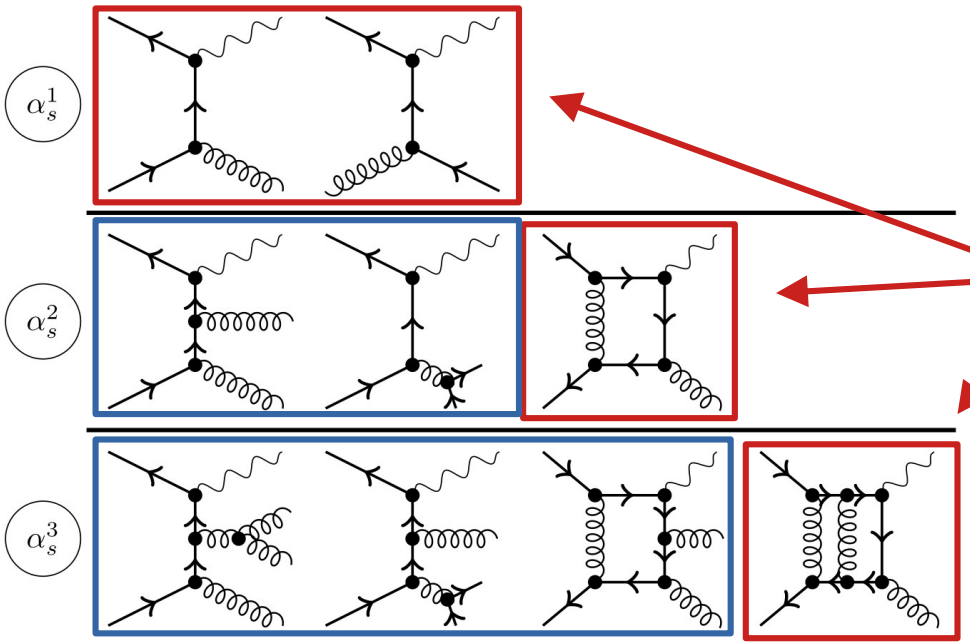


high-pT boson as a tag

In our case: isolated photon (hybrid cone isolation) with $p_{T,\gamma} > 500\ \mathrm{GeV}\ ,\ |\eta_\gamma| < 2.4$

Calculational framework: fixed-order

$$pp \to \gamma + X$$



perturbative expansion:

$$\frac{d\sigma^{[N]}}{dR_L} = \frac{d\sigma^{[N](0)}}{dR_L} + \frac{d\sigma^{[N](1)}}{dR_L} + \frac{d\sigma^{[N](2)}}{dR_L} + \mathcal{O}\left(\alpha_s^4\right)$$

$$\sim \delta(R_L)$$

differential R_L distributions starts at α_s^2

→ effectively renders NNLO to NLO

Implemented in C++ sector-improved residue subtraction framework [1408.2500][1911.00479]

Calculational framework: all order structure

Logarithmic structure in the small R_L limit:

$$\frac{d\hat{\sigma}^{[N]}}{dR_L} = \sum_{L=1}^{\infty} \sum_{j=-1}^{L-1} \left(\frac{\alpha_s}{4\pi}\right)^L e_{L,j} \mathcal{L}^j(R_L)\,,$$

$$\text{where} \quad \mathcal{L}^{-1}(R_L) = \delta(R_L),\ \mathcal{L}^j(R_L) = \left[\ln^j(R_L)/R_L\right]_+$$

Collinear factorization theorem
[1905.01310][2004.11381]

$$\Sigma^{[N]}_{e^+e^-}(x_L) = \int_0^1 dx\, x^N \vec{J}^{[N]} \left(\ln \frac{x_L x^2 Q^2}{\mu^2}, \mu\right) \cdot \vec{H}_{e^+e^-} \left(x, \ln \frac{Q^2}{\mu^2}, \mu\right)$$

ENC jet function

Hard function
(single inclusive fragmentation function)

Calculational framework: collinear factorization

At hadron colliders:

$$\frac{d\sigma^{[N]}}{dR_L} = \frac{d}{dR_L}\Bigg[\sum_{ab}\iint_0^1 \mathrm{d}x_1\mathrm{d}x_2\; f_a(x_1,\mu_F)\, f_b(x_2,\mu_F) \times \sum_{c=q,g}\int \mathrm{d}z\, z^N \times d\hat{\sigma}_{ab\to c\gamma+X}(x_1,x_2,z,\mu_F,\mu_R,\mu_J,\alpha_s(\mu_R)) \times J_c^{[N]}\left(\ln\frac{R_L^2 z^2 p_{\gamma,T}^2}{\mu_J^2}, \ln\frac{\mu_R}{\mu_J}, \alpha_s(\mu_R)\right)\Bigg]$$

RGE evolution of ENC jet function:

$$\frac{d}{d\ln\mu^2}\vec{J}^{[N]}\left(\ln\frac{R_L^2 z^2 p_{\gamma,T}^2}{\mu^2}, \alpha_s(\mu)\right) = \int_0^1 dy\, y^N \vec{J}^{[N]}\left(\ln\frac{R_L^2 y^2 z^2 p_{\gamma,T}^2}{\mu^2}, \alpha_s(\mu)\right)\cdot \widehat{P}(y,\alpha_s(\mu))$$

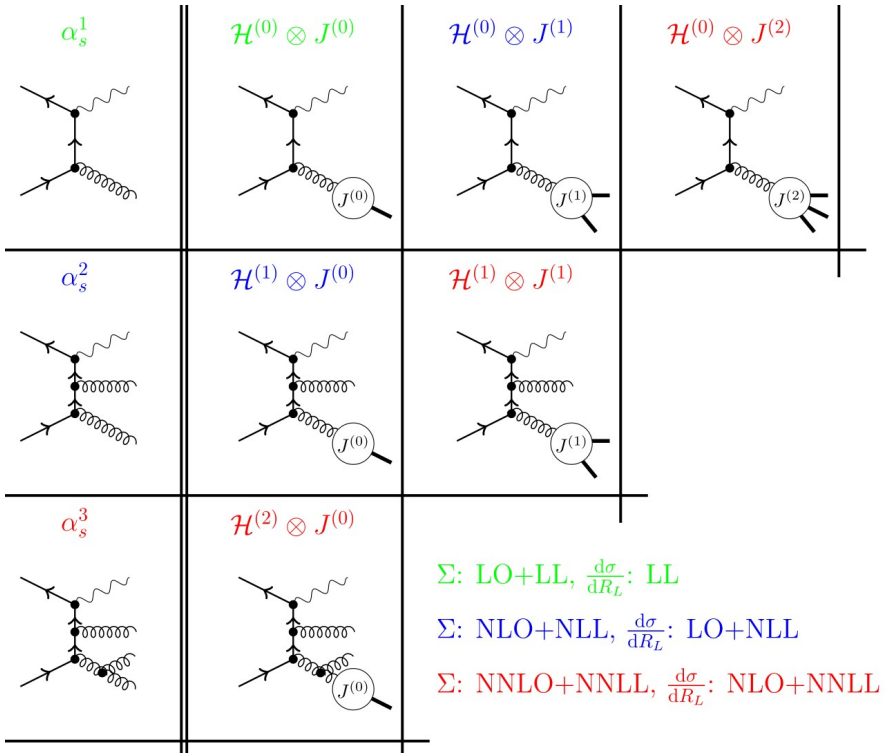
Solving this by an ansatz:

$$\vec{J}^{[N]}\left(\ln\frac{R_L^2 z^2 p_{\gamma,T}^2}{\mu^2}, \alpha_s(\mu)\right) = \sum_{n=0}^{\infty}\sum_{m=0}^{n}\left(\frac{\alpha_s(\mu)}{4\pi}\right)^n \frac{\vec{j}_{n,m}^{[N]}}{m!}\ln^m\frac{R_L^2 z^2 p_{\gamma,T}^2}{\mu^2}$$

→ needs boundary terms from fixed order $\vec{j}_{n,0}^{[N]}$
→ known at NLO for general N
→ known at NNLO up to N = 6

[2004.11381][2307.07510][2606.02714]

Fixed-order matching with resummation



$$\frac{d\hat{\sigma}^{[N]}_{\text{match}}}{dR_L} = \frac{d\hat{\sigma}^{[N]}_{\text{resum}}}{dR_L} + \frac{d\hat{\sigma}^{[N]}_{\text{ns}}}{dR_L}$$

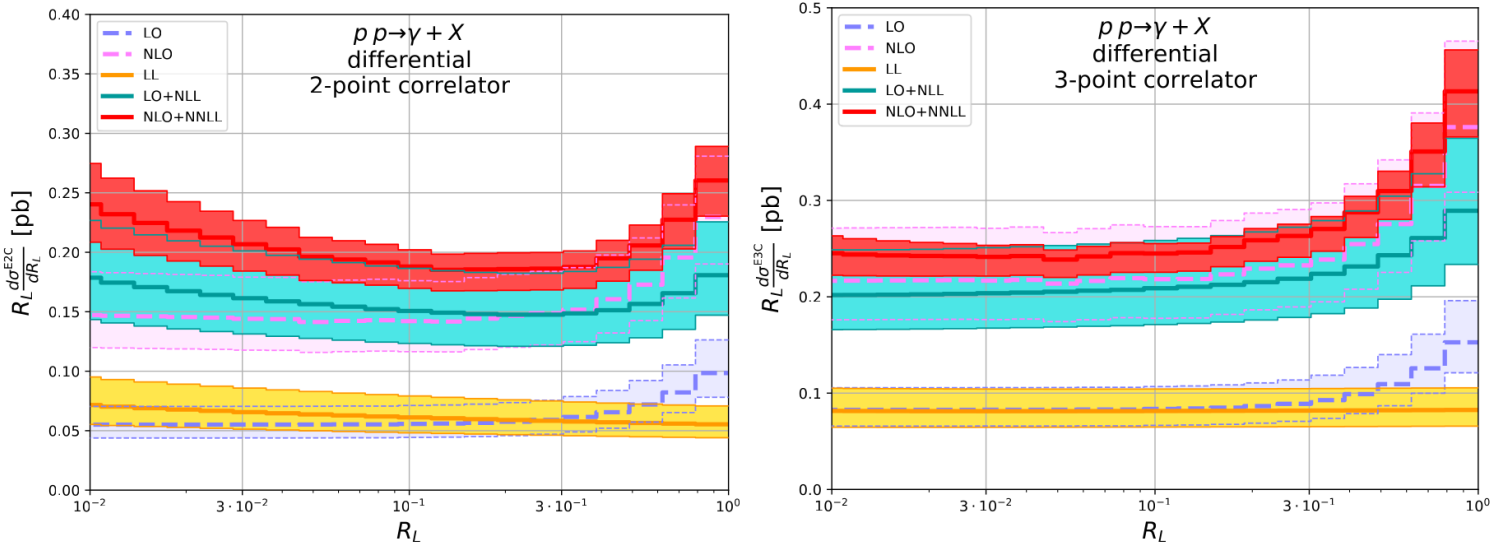
N^{k}LL:
→ (k+1) loop splitting functions
→ (k) order hard + jet function
→ (k+1) loop beta function

Combination of hard+jet function:

$$\int d\sigma_{\gamma\, i+X} \left(\frac{p_{i,T}}{p_{\gamma,T}}\right)^N \alpha_s^l(\mu_R) \ln^m\left(\frac{p_{i,T}^2}{\mu_J^2}\right) \ln^n\left(\frac{\mu_R^2}{\mu_J^2}\right)$$

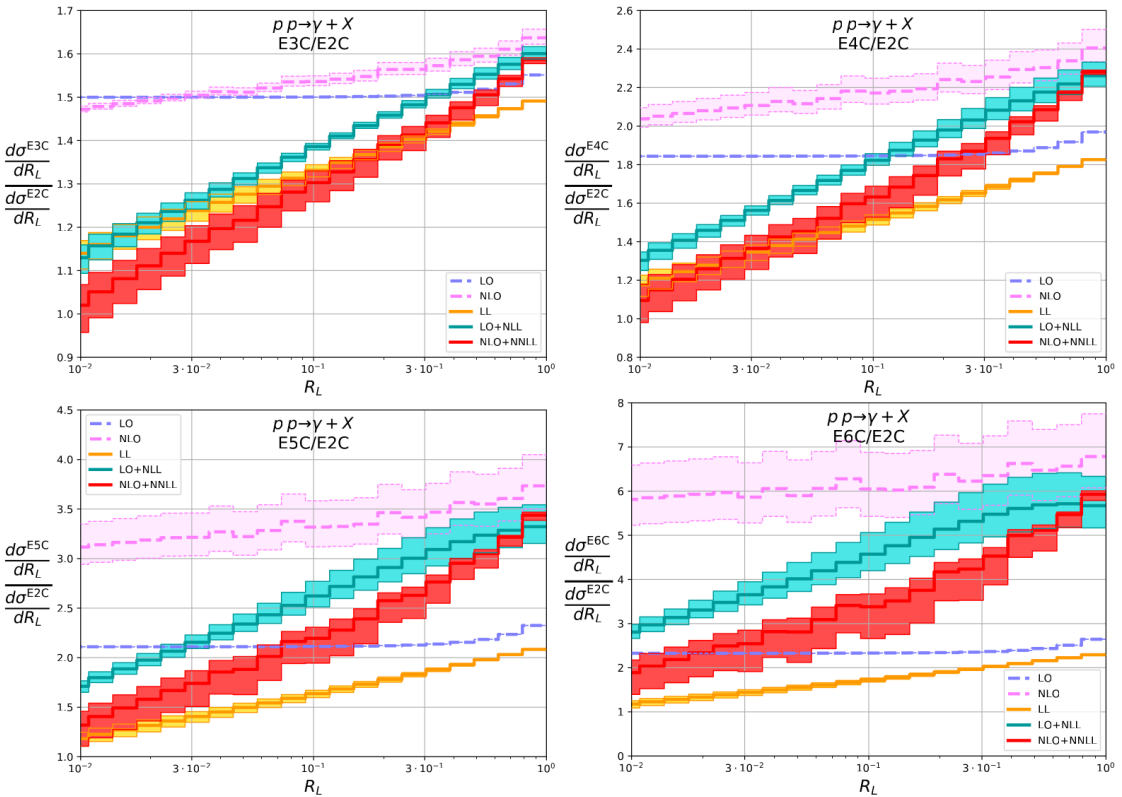
Single inclusive fragmentation implementation with $z = p_{i,T}/p_{\gamma,T}$
[2102.08267][2411.09684][2503.11489][2503.21866]

Phenomenology



Large perturbative corrections from NLO QCD
→ NLL to NNLL shows convergence

ENC ratios



$$\frac{d\sigma^{[N]}/dR_L}{d\sigma^{[2]}/dR_L}$$

→ certain experimental uncertainties cancel (e.g. luminosity)

→ partial cancellation of non-perturbative effects

→ higher sensitivity to α_s

Non perturbative effects

$$\frac{d\sigma^{[N]}}{dR_L} = \frac{d\hat{\sigma}^{[N]}_{\text{match}}}{dR_L} + \frac{d\sigma^{[N]}_{\text{NP}}}{dR_L},$$

Leading non-perturbative correction:

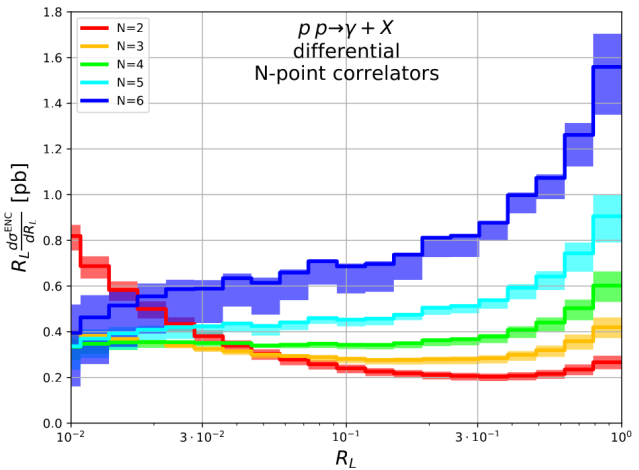
[hep-ph/9902341][2405.19396][2406.06668] $\mathcal{O}\left(\frac{\Lambda_{QCD}}{p_T}\right)$

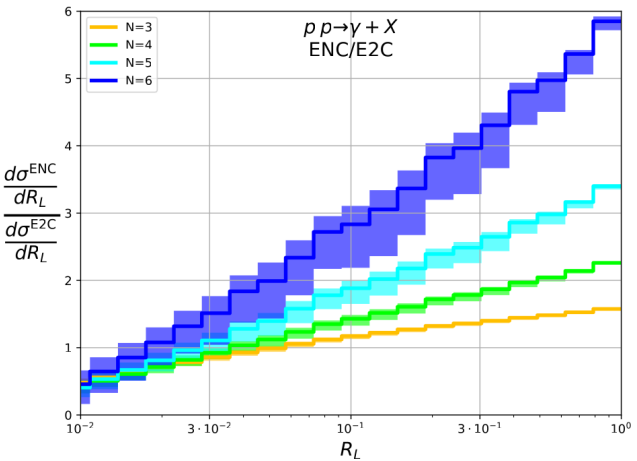
$$\Omega_{1\kappa} \equiv \frac{1}{N_\kappa}\langle 0|\text{tr}\left[\overline{Y}^{\dagger\kappa}_{\bar{n}} Y^{\dagger\kappa}_{n} \mathcal{E}_T(0) Y^{\kappa}_{n} \overline{Y}^{\kappa}_{\bar{n}}\right]|0\rangle$$

Incorporate them into the jet function:

$$J^{\kappa[N]}\left(\ln\frac{R_L^2 x^2 p_T^2}{\mu^2}, \mu\right) = \hat{J}^{\kappa[N]}\left(\ln\frac{R_L^2 x^2 p_T^2}{\mu^2}, \mu\right) - \frac{N}{x\, p_T R_L} \hat{J}^{\kappa[N-1]}_{\text{NP}}\left(\ln\frac{R_L^2 x^2 p_T^2}{\mu^2}, \mu\right)$$

$$\hat{J}^{\kappa[N-1]}_{\text{NP}}\left(\ln\frac{R_L^2 p_T^2}{(\mu_J^0)^2}, \mu_J^0\right) = \overline{\Omega}_{1\kappa} + \mathcal{O}(\alpha_s)$$



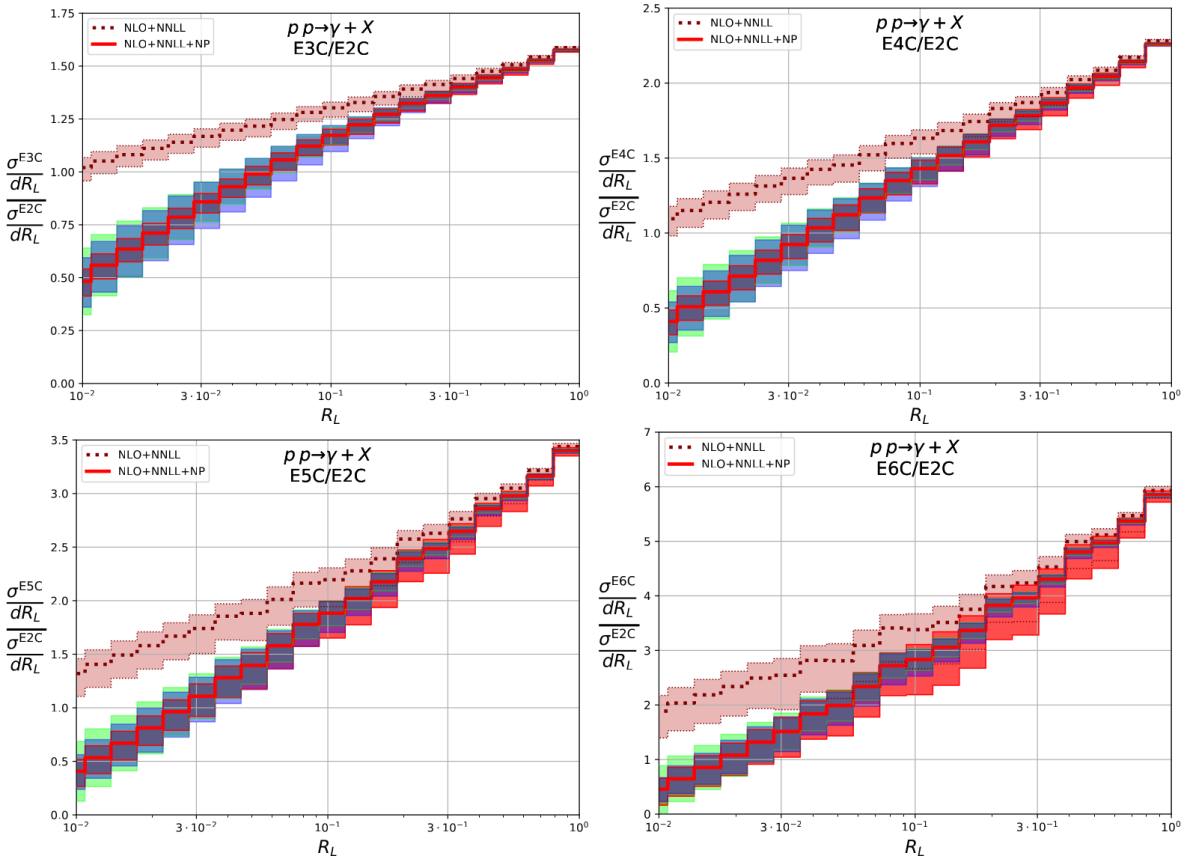


From data: $\overline{\Omega}_{1q} = 0.305\ \text{GeV}$

$$\overline{\Omega}_{1g} = C_A/C_F\, \overline{\Omega}_{1q}$$

not measured
→ treat as additional unc.

Final results for the ratios



(red band)
Perturbative uncertainties (scale var.)

(green)
Unknown higher order constant
of NP jet function

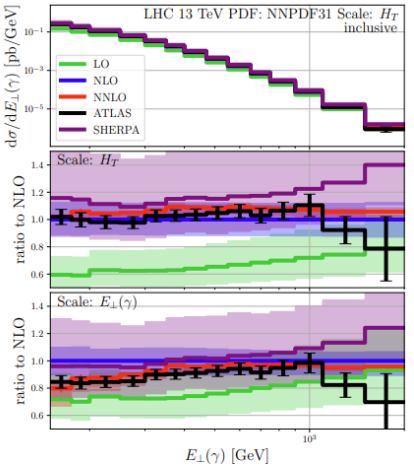
(blue)
uncertainties from unknown $\bar{\Omega}_{1,g}$

Outlook: towards NNLO QCD

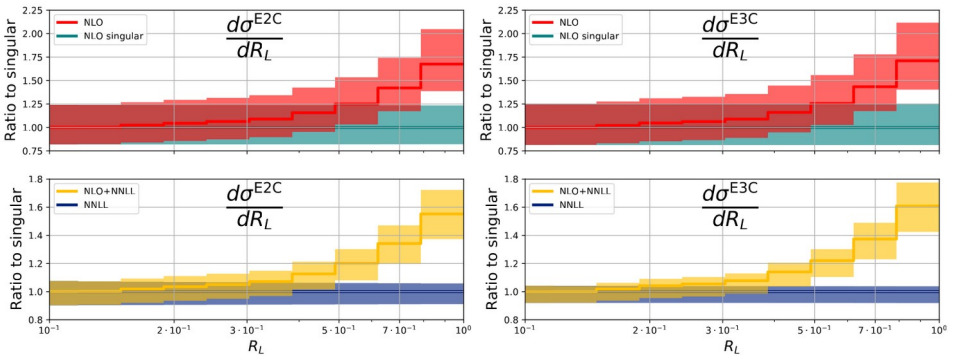
Ingredient: NNLO QCD for photon + two partons

→ all amplitudes and subtractions are known [2304.06682]

→ for the differential and ratios no N3LO subtraction is needed



Non-singular contribution only sizeable above $R_L > 0.1$:
→ NNLO QCD should be feasible



Take-home messages

- (projected) **Energy-energy correlators** provide a **sensitive probe** to QCD kinematics
- **V+X provides a jet independent laboratory** at hadron colliders
- **Precision predictions** based on collinear factorization available